

PVsyst - Simulation report

Grid-Connected System

Project: **BATTERY (4)**

Variant: **15.12**

No 3D scene defined, no shadings

System power: **15.12** kWp

Kampung Chuang - Malaysia

Author

NORTHERN SOLAR SDN. BHD. (Malaysia)



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PVsyst V8.1.3

VC0, Simulation date:
with V8.1.3

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Project summary

Geographical Site

Kampung Chuang

Malaysia

Situation

Latitude **3.4869** °(N)

Longitude **101.6284** °(E)

Altitude 22 m

Time zone UTC+8

Project settings

Albedo 0.20

Weather data

Kampung Chuang

Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

Simulation for year no 1

No 3D scene defined, no shadings

Orientation #1

Fixed plane

Tilt/Azimuth **20 / 90** °

Orientation #4

Fixed plane

Tilt/Azimuth **20 / 90** °

System information

PV Array

Nb. of modules **24** units

Pnom total **15.12** kWp

Orientation #2

Fixed plane

Tilt/Azimuth **20 / 90** °

Near Shadings

no Shadings

Inverters

Nb. of units **1** unit

Total power **15** kWac

Pnom ratio **1.01**

Orientation #3

Fixed plane

Tilt/Azimuth **20 / 90** °

User's needs

Monthly values

Battery pack

Storage strategy: Self-consumption

Nb. of units **1** unit

Voltage 51 V

Capacity **314** Ah

Results summary

Simulation step Hourly

Produced Energy **21174** kWh/year

Used Energy **10438** kWh/year

Specific production **1400** kWh/kWp/year

Perf. Ratio PR **81.40** %

Solar Fraction SF **84.76** %

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General parameters**Grid-Connected System****No 3D scene defined, no shadings****Simulation step**

Hourly

Orientation #1**Fixed plane**Tilt/Azimuth **20 / 90 °****Orientation #2****Fixed plane**Tilt/Azimuth **20 / -90 °****Orientation #3****Fixed plane**Tilt/Azimuth **20 / 0 °****Orientation #4****Fixed plane**Tilt/Azimuth **20 / 180 °****Models used**Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate**Horizon**Average Height **2.6 °****Near Shadings**

no Shadings

Storage

Kind Self-consumption

User's needs

Monthly values

Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.	Year	
													kWh/mth

PV Array Characteristics**PV module**Manufacturer **Astronergy**Model **CHSM66RN(DG)/F-BH-630**

(Custom parameters definition)

CHSM66RN(DG)F-BH-630.PANUnit Nom. Power **630 Wp****Inverter**Manufacturer **Hoymiles**Model **HIT-15L-G3**

(Custom parameters definition)

Unit Nom. Power **15.0 kWac****Array #1 - PV Array**Orientation **1**Tilt/Azimuth **20/90 °**Number of PV modules **6 units**Nominal (STC) **3780 Wp**Modules **1 unit x 6 In series****At operating cond. (50°C)**Pmpp **3528 Wp**U mpp **226 V**I mpp **16 A**Number of inverters **1 * MPPT 25% 0.3 unit**Total power **3.8 kWac**Operating voltage **150-900 V**Pnom ratio (DC:AC) **1.01****Array #2 - Sub-array #2**Orientation **2**Tilt/Azimuth **20/-90 °**Number of PV modules **6 units**Nominal (STC) **3780 Wp**Modules **1 unit x 6 In series****At operating cond. (50°C)**Pmpp **3528 Wp**U mpp **226 V**I mpp **16 A**Number of inverters **1 * MPPT 25% 0.3 unit**Total power **3.8 kWac**Operating voltage **150-900 V**Pnom ratio (DC:AC) **1.01**

**PV Array Characteristics****Array #3 - Sub-array #3**

Orientation	3		
Tilt/Azimuth	20/0 °		
Number of PV modules	6 units	Number of inverters	1 * MPPT 25% 0.3 unit
Nominal (STC)	3780 Wp	Total power	3.8 kWac
Modules	1 unit x 6 In series		
At operating cond. (50°C)		Operating voltage	150-900 V
Pmpp	3528 Wp	Pnom ratio (DC:AC)	1.01
U mpp	226 V		
I mpp	16 A		

Array #4 - Sub-array #4

Orientation	4		
Tilt/Azimuth	20/180 °		
Number of PV modules	6 units	Number of inverters	1 * MPPT 25% 0.3 unit
Nominal (STC)	3780 Wp	Total power	3.8 kWac
Modules	1 unit x 6 In series		
At operating cond. (50°C)		Operating voltage	150-900 V
Pmpp	3528 Wp	Pnom ratio (DC:AC)	1.01
U mpp	226 V		
I mpp	16 A		

Total PV power

Nominal (STC)	15 kWp
Total	24 modules
Module area	64.8 m²
Cell area	60.5 m²

Total inverter power

Total power	15 kWac
Number of inverters	1 unit
Pnom ratio	1.01
No power sharing	

Grid storage - Self-consumption**Battery**

Manufacturer	Hoymiles
Model	LB-16D-G3

Battery pack

Nb. of units	1 unit
Voltage	51 V
Nominal Capacity (C10)	314 Ah
Nominal energy	16.1 kWh
Operating SOC limits	95.0 / 20.0 %
Useable energy	12.1 kWh
Operating temperature	Fixed 20 °C

Battery input charger

Model	Generic
Max. charging power	8.00 kWdc
Max./Euro effic.	98.0 / 96.5 %

Battery to Grid inverter

Model	Generic
Max. discharging power	1.35 kWac
Max./Euro effic.	98.0 / 96.5 %

Array losses**Array Soiling Losses**

Loss Fraction	1.5 %
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Thermal Loss factor

Module temperature according to irradiance	
Uc (const)	20.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

LID - Light Induced Degradation

Loss Fraction	1.45 %
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Module Quality Loss

Loss Fraction	-0.75 %
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Module mismatch losses

Loss Fraction	1.00 % at MPP
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Module average degradation

Year no	1
Loss factor	0.55 %/year
Imp / Vmp contributions	80% / 20%
Mismatch due to degradation	
Imp RMS dispersion	0.4 %/year
Vmp RMS dispersion	0.4 %/year



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Array losses

IAM loss factor

ASHRAE Param.: IAM = $1 - b_o (1/\cos i - 1)$

bo Param. 0.05

DC wiring losses

Global wiring resistance 10 mΩ
Loss Fraction 1.71 % at STC

Array #1 - PV Array

Global array res. 471 mΩ
Loss Fraction 3.00 % at STC

Array #3 - Sub-array #3

Global array res. 236 mΩ
Loss Fraction 1.50 % at STC

Array #2 - Sub-array #2

Global array res. 236 mΩ
Loss Fraction 1.50 % at STC

Array #4 - Sub-array #4

Global array res. 236 mΩ
Loss Fraction 1.50 % at STC

System losses

Unavailability of the system

Time fraction 1.0 %
3.7 days,
3 periods

AC wiring losses

Inv. output line up to injection point

Inverter voltage 400 Vac tri
Loss Fraction 1.00 % at STC

Global System

Wire section Copper 3 x 3 mm²
Wires length 14 m



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Main results

Simulation step

Hourly

System Production

Produced Energy **21174** kWh/year

Used Energy **10438** kWh/year

Specific production

1400 kWh/kWp/year

Perf. Ratio PR

81.40 %

Solar Fraction SF

84.76 %

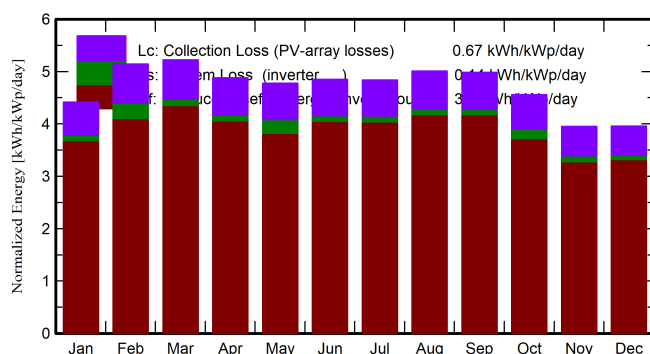
Battery ageing (State of Wear)

Cycles SOW 65.6 %

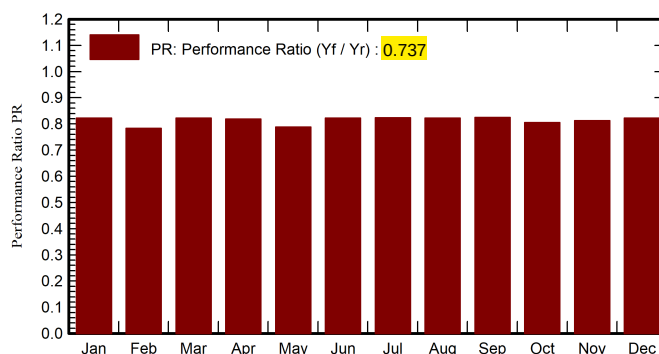
Static SOW 90.0 %

Battery lifetime 2.9 years

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m ²	DiffHor kWh/m ²	T_Amb °C	GlobInc kWh/m ²	GlobEff kWh/m ²	EArray kWh	E_User kWh	E_Solar kWh	E_Grid kWh	EFrGrid kWh
January										
February										
March										
April										
May										
June										
July										
August										
September										
October										
November										
December										
Year										

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

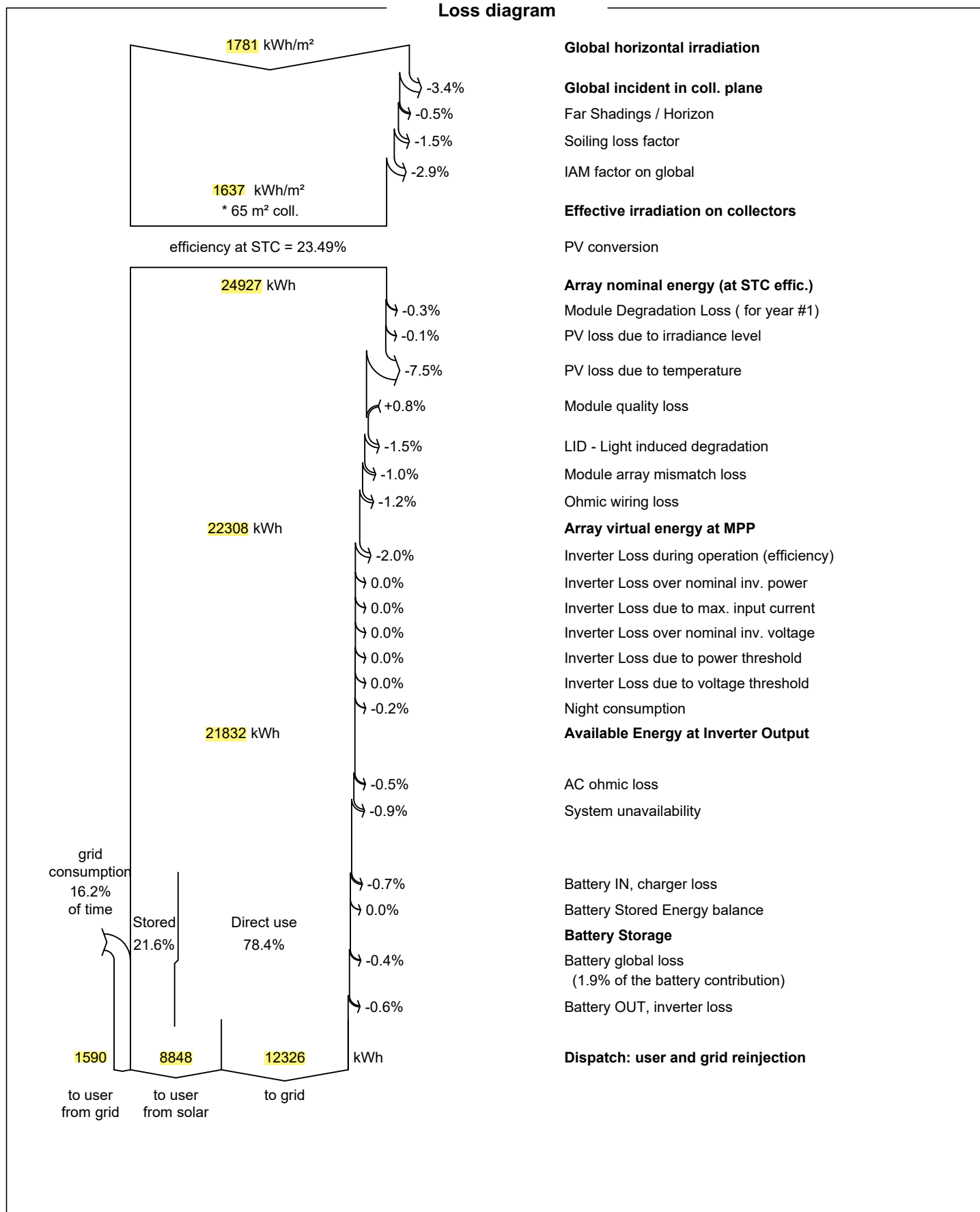
EArray Effective energy at the output of the array

E_User Energy supplied to the user

E_Solar Energy from the sun

E_Grid Energy injected into grid

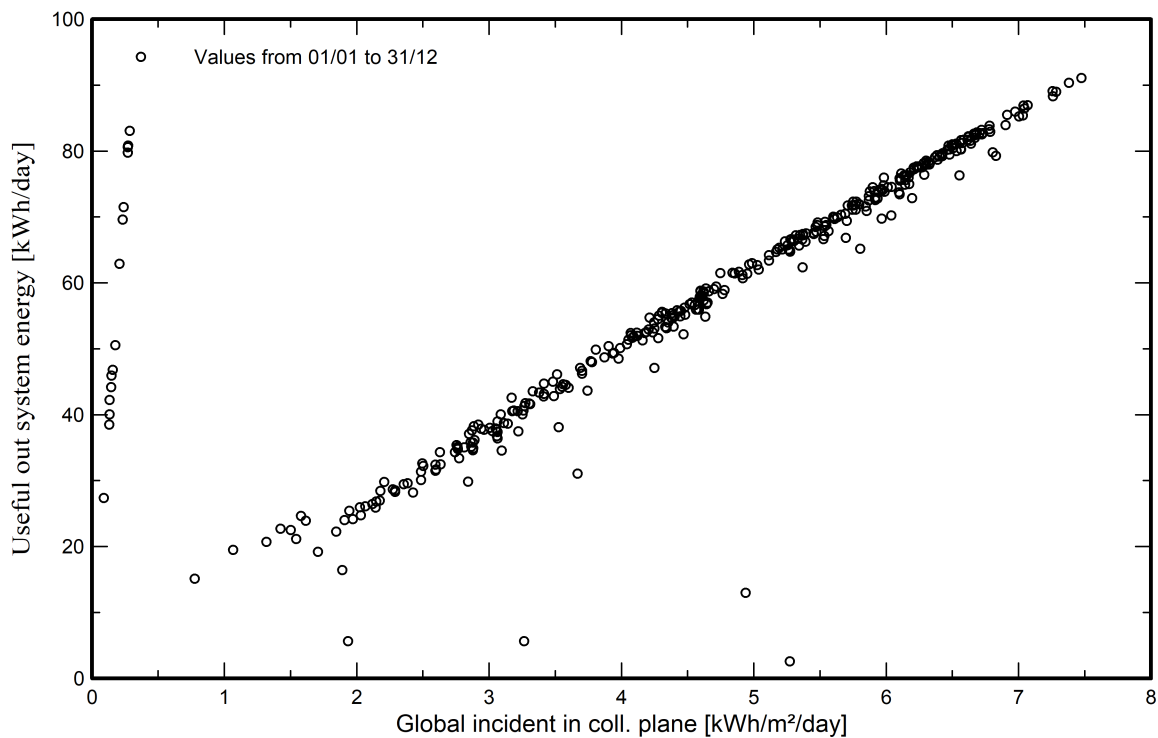
EFrGrid Energy from the grid

**Loss diagram**

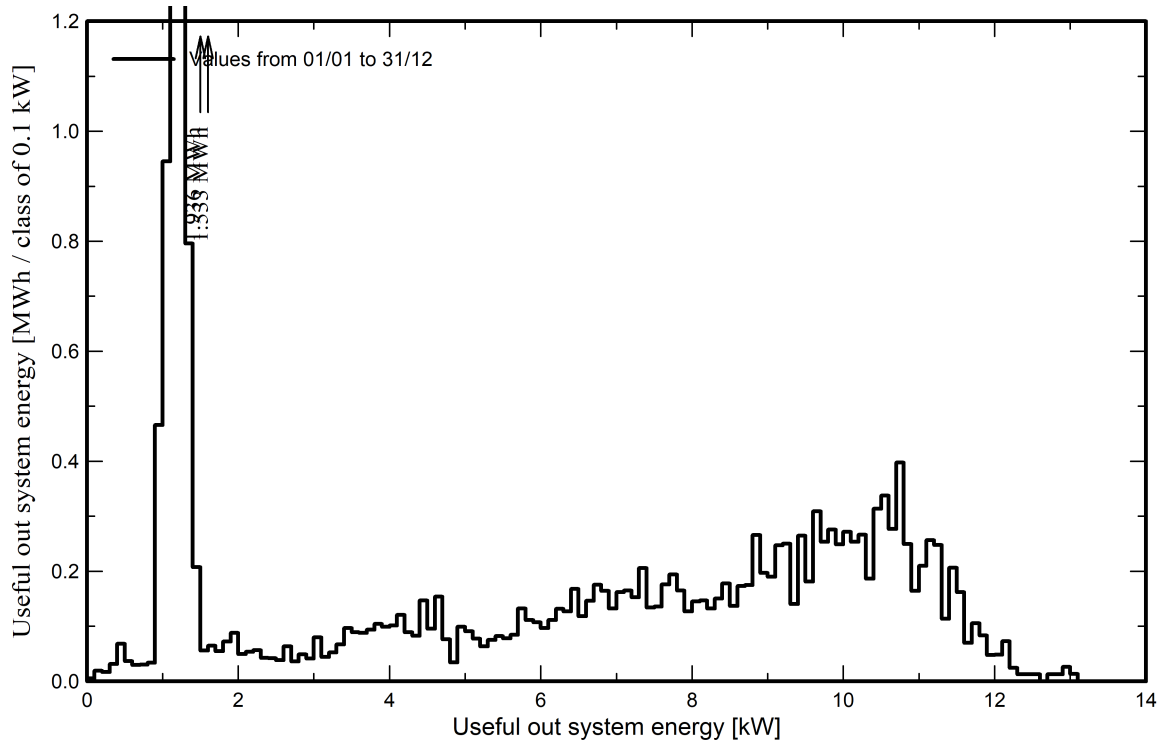


Predef. graphs

Daily Input/Output diagram



System Output Power Distribution





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P50 - P90 evaluation

Weather data

Source	Meteonorm 9.0 dll, Sat=100%
Kind	Specific year
Year	Synthetic
Year-to-year variability(Variance)	4.0 %
Specified Deviation	
Year deviation from average	0.0 %

Global variability (weather data + system)

Variability (Quadratic sum)	5.9 %
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Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %
Custom variability	4.0 %

Annual production probability

Variability	666 kWh
P50	11211 kWh
P90	10367 kWh
P75	10772 kWh

Probability distribution

